

Research Statement

Interpretable and Robust Machine Learning for Physics-Based Numerical Simulations

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1 Introduction and Motivation

The integration of Machine Learning (ML) into physical sciences has shifted from simple data-fitting to the creation of high-fidelity surrogate models capable of accelerating computationally expensive numerical solvers. However, the adoption of these models in critical industrial Digital Twins is hindered by two primary challenges: the "black-box" nature of deep learning architectures and their fragility when faced with out-of-distribution (OOD) data. This research statement outlines my proposal to address these limitations through advanced statistical methods, ensuring that hybrid physics-data models are both interpretable and robust.

2 Current Research: Physics-Informed Digital Twins

In my current work at Sonatrach, I have developed a generative surrogate model for gas turbine monitoring using **Mamba-SSM** and **xLSTM** architectures. The core of my contribution is the design of custom **Physics-Informed Neural Network (PINN)** loss functions that enforce thermodynamic consistency (e.g., energy conservation, pressure-temperature relationships). This work has demonstrated that while physical constraints improve generalization, they do not inherently provide a measure of model reliability. My experience with these hybrid models has revealed the critical need for **Uncertainty Quantification (UQ)** to detect when a surrogate is operating outside its domain of validity.

3 Proposed Research Directions

Building upon the objectives of the **Catalyst (Reliable Hybrid Model Forge)** project, I propose to investigate the following three axes:

3.1 1. Global Sensitivity Analysis for Neural Surrogates

I intend to use Global Sensitivity Analysis (GSA) to identify which input parameters or internal model components (e.g., specific layers or weights) most significantly influence the output in physics-based tasks. By quantifying these influences, we can move towards a "white-box" understanding of how deep learning models emulate physical phenomena, facilitating better alignment with established numerical laws.

3.2 2. Optimal Transport for Generalization Assessment

Generalization in physics-based ML is often a matter of how well the model maps the training distribution to the test distribution. I propose to leverage **Optimal Transport (OT)** methods to analyze the geometry of learned representations. By measuring the distance between the training manifold and new simulation contexts, we can establish a quantitative threshold for model reuse, preventing catastrophic failures in unseen physical regimes.

3.3 3. Statistical Depth for Out-of-Distribution Detection

To enhance model robustness, I will investigate **Statistical Depth** metrics. These tools allow for the identification of "central" vs. "outlying" data points in high-dimensional spaces. In the context of a Digital Twin, this means the model will be capable of signaling a "low-confidence" warning when the input physics parameters represent a scenario (e.g., an extreme fault condition) that was not adequately covered during training.

4 Alignment with the Catalyst Project

My background in production-grade PyTorch implementation, combined with my rigorous training in statistics and numerical methods, aligns perfectly with the interdisciplinary nature of the **EDT program**. I am particularly interested in contributing to the ****Artemis platform****, ensuring that the theoretical breakthroughs in interpretability are translated into usable tools for the digital twin community.

5 Conclusion

The goal of my PhD will be to bridge the gap between the predictive power of state-of-the-art ML and the reliability of classical numerical analysis. By developing a framework for interpretable and robust physics-aware AI, I aim to provide the foundational tools necessary for the next generation of trustworthy industrial Digital Twins.